P8106_Final

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(tidymodels)
```

```
## -- Attaching packages --------------------------------------- tidymodels 1.4.1 --
## v broom        1.0.9     v rsample      1.3.2
## v dials        1.4.2     v tailor       0.1.0
## v infer        1.0.9     v tune         2.0.1
## v modeldata    1.5.1     v workflows    1.3.0
## v parsnip      1.4.1     v workflowsets 1.1.1
## v recipes      1.3.1     v yardstick    1.3.2

## Warning: package 'parsnip' was built under R version 4.5.2

## Warning: package 'rsample' was built under R version 4.5.2

## -- Conflicts ------------------------------------------ tidymodels_conflicts() --
## x rsample::calibration()   masks caret::calibration()
```

```
## x scales::discard()        masks purrr::discard()
## x dplyr::filter()          masks stats::filter()
## x recipes::fixed()         masks stringr::fixed()
## x dplyr::lag()             masks stats::lag()
## x caret::lift()            masks purrr::lift()
## x yardstick::precision()   masks caret::precision()
## x yardstick::recall()      masks caret::recall()
## x yardstick::sensitivity() masks caret::sensitivity()
## x yardstick::spec()        masks readr::spec()
## x yardstick::specificity() masks caret::specificity()
## x recipes::step()          masks stats::step()
```

```
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2
```

```
## Loaded gbm 2.2.3
## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.c
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:tune':
##
##     tune
##
## The following object is masked from 'package:rsample':
##
##     permutations
##
## The following object is masked from 'package:parsnip':
##
##     tune
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(pROC)
```

```
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
library(patchwork)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
library(ggplot2)
library(DALEX)
```

```
## Welcome to DALEX (version: 2.5.3).
## Find examples and detailed introduction at: http://ema.drwhy.ai/
##
##
## Attaching package: 'DALEX'
##
## The following object is masked from 'package:dplyr':
##
##     explain
```

Load the data

```r
load("severity.RData")
dat <- na.omit(dat)
cat(
  "Number of observations removed due to N/A:",
  sum(!complete.cases(dat))
)
```

```
## Number of observations removed due to N/A: 0
```

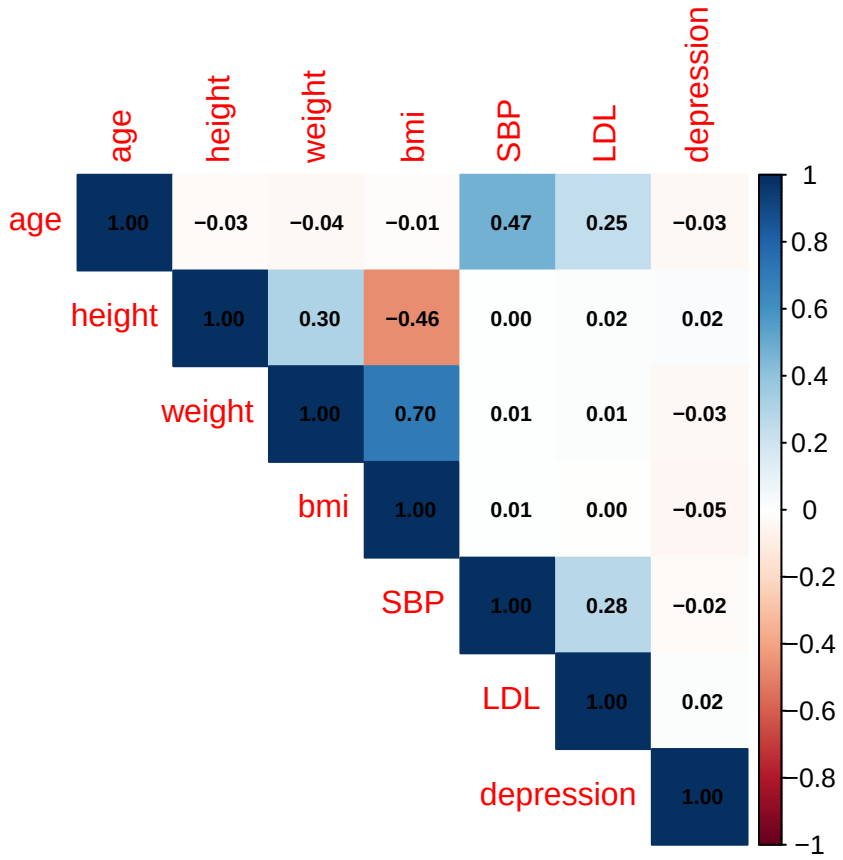
```r
set.seed(4027)
my_dat <- dat[sample(nrow(dat), 1000), ]
```

```r
# Correlations
continuous_vars <- my_dat %>%
  select(
    age,
    height,
    weight,
    bmi,
    SBP,
    LDL,
    depression
  )

cor_matrix <- cor(continuous_vars)

corrplot(
  cor_matrix,
```

```
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 0.7
)
```




```
my_dat <- my_dat %>%
  select(-id, -height, -weight)
```

Convert Variables

```
my_dat <- my_dat %>%
  mutate(
    severity = factor(severity,
                      levels = c(0,1),
                      labels = c("NotSevere","Severe")),

    gender = factor(gender,
                    levels = c(0,1),
                    labels = c("Female","Male")),

    hypertension = factor(hypertension,
```

```
                    levels = c(0,1),
                    labels = c("No","Yes")),
    diabetes = factor(diabetes,
                levels = c(0,1),
                labels = c("No","Yes")),
    vaccine = factor(vaccine,
              levels = c(0,1),
              labels = c("Not vaccinated","Vaccinated")),

    race = factor(race,
                  levels = c(1,2,3,4),
                  labels = c("White","Asian","Black","Hispanic")),

    smoking = factor(smoking,
                     levels = c(0,1,2),
                     labels = c("Never","Former","Current"))
  )
```

Train test split

```
set.seed(1234)
split <- initial_split(my_dat,
                       prop = 0.8,
                       strata = severity)

train_dat <- training(split)
test_dat  <- testing(split)
```

Exploratory Analysis

```
# Summary table
summary_table <- train_dat %>%
  summary()
summary_table
```

```
##       age           gender          race        smoking         bmi
##  Min.   :48.00   Female:408   White   :491   Never  :469   Min.   :20.5
##  1st Qu.:57.00   Male  :391   Asian   : 46   Former :242   1st Qu.:26.0
##  Median :60.00                Black   :172   Current: 88   Median :27.6
##  Mean   :60.17                Hispanic: 90                 Mean   :27.8
##  3rd Qu.:63.00                                             3rd Qu.:29.5
##  Max.   :74.00                                             Max.   :36.8
##  diabetes  hypertension      SBP             LDL                 vaccine
##  No :688   No :408      Min.   :101.0   Min.   : 52.0   Not vaccinated:303
##  Yes:111   Yes:391      1st Qu.:124.0   1st Qu.: 96.0   Vaccinated    :496
##                         Median :130.0   Median :111.0
##                         Mean   :130.1   Mean   :109.9
```

```
##                          3rd Qu.:136.0   3rd Qu.:123.0
##                          Max.   :159.0   Max.   :178.0
##    depression          severity
##  Min.   : 0.000   NotSevere:546
##  1st Qu.: 6.000   Severe   :253
##  Median : 7.000
##  Mean   : 6.996
##  3rd Qu.: 8.000
##  Max.   :13.000
```

```
# Continuous variable
p1 <- ggplot(my_dat,
             aes(x = severity,
                 y = age,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Age by Severity")

p2 <- ggplot(my_dat,
             aes(x = severity,
                 y = bmi,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("BMI by Severity")

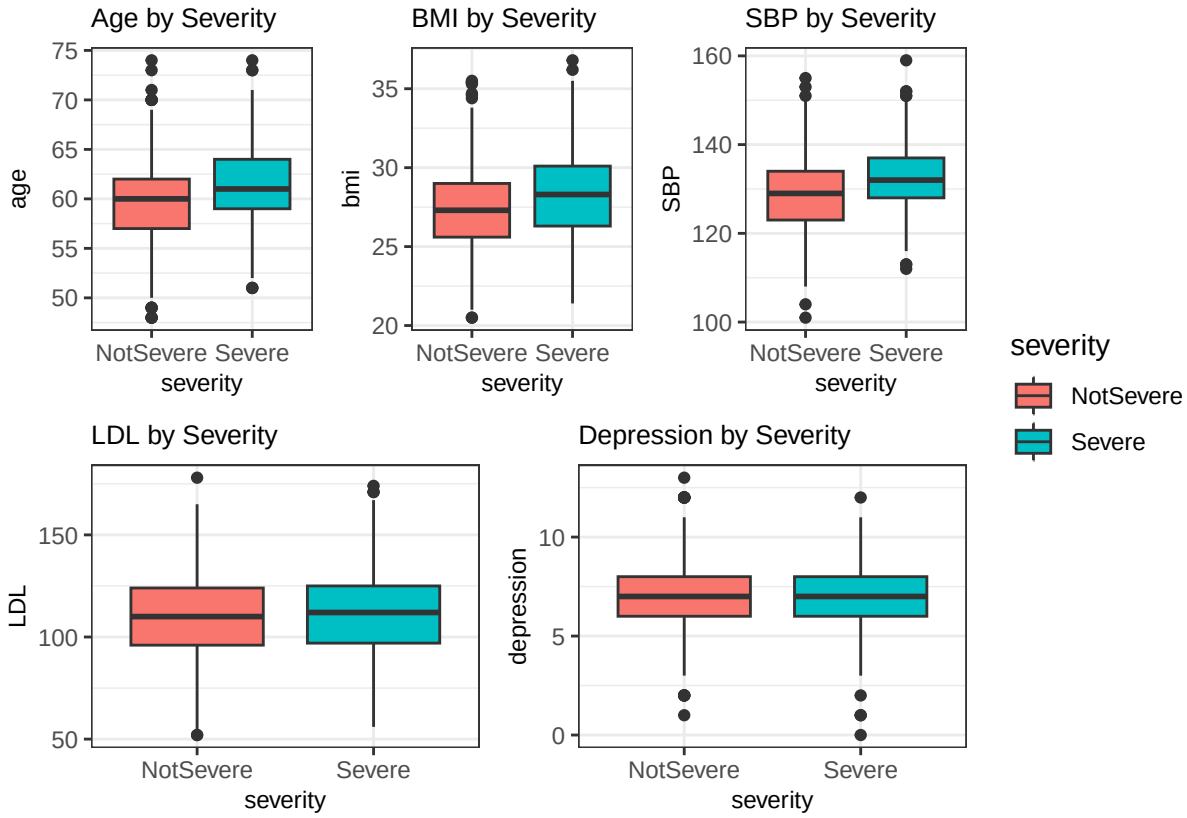
p3 <- ggplot(my_dat,
             aes(x = severity,
                 y = SBP,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("SBP by Severity")

p4 <- ggplot(my_dat,
             aes(x = severity,
                 y = LDL,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("LDL by Severity")

p5 <- ggplot(my_dat,
             aes(x = severity,
                 y = depression,
                 fill = severity)) +
  geom_boxplot() +
  theme_bw() +
  ggtitle("Depression by Severity")

(p1 + p2 + p3) /
 (p4 + p5) +
plot_layout(guides = "collect") &
```

```
theme(
  plot.title = element_text(size = 10),
  axis.title = element_text(size = 9),
  axis.text = element_text(size = 9)
)
```




```
# Categorical variables
c1 <- ggplot(my_dat,
             aes(x = gender,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Gender vs Severity")

c2 <- ggplot(my_dat,
             aes(x = race,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Race vs Severity")

c3 <- ggplot(my_dat,
             aes(x = smoking,
                 fill = severity)) +
  geom_bar(position = "fill") +
```

```
  theme_bw() +
  ggtitle("Smoking vs Severity")

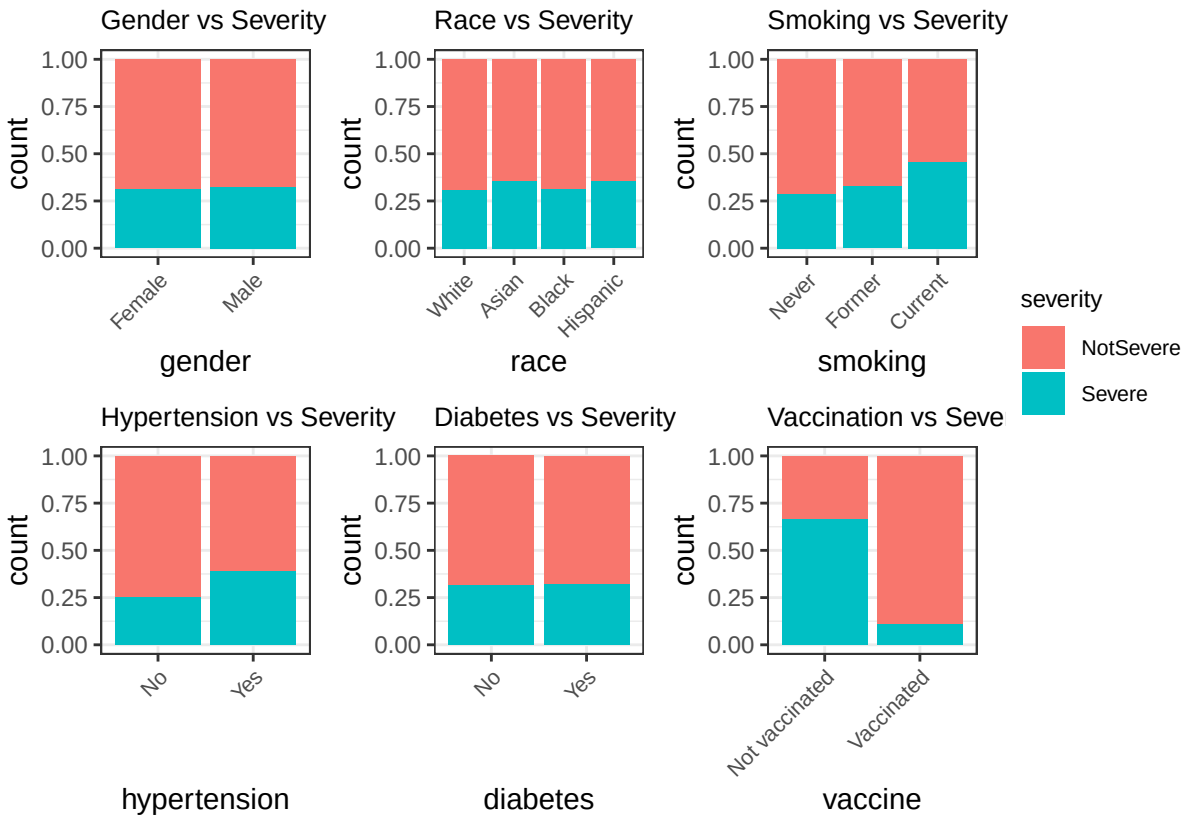
c4 <- ggplot(my_dat,
             aes(x = hypertension,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Hypertension vs Severity")


c5 <- ggplot(my_dat,
             aes(x = diabetes,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Diabetes vs Severity")


c6 <- ggplot(my_dat,
             aes(x = vaccine,
                 fill = severity)) +
  geom_bar(position = "fill") +
  theme_bw() +
  ggtitle("Vaccination vs Severity")

common_theme <- theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45,
                               hjust = 1,
                               size = 8),
    plot.title = element_text(size = 10),
    legend.title = element_text(size = 9),
    legend.text = element_text(size = 8)
  )

((c1 + c2 + c3) /
(c4 + c5 + c6)) +
  plot_layout(guides = "collect") &
  common_theme
```



Logistic Regression

```
logit_fit <- glm(
  severity ~ age + gender + race + smoking +
    bmi + hypertension + diabetes +
    SBP + LDL + vaccine + depression,
  data = train_dat,
  family = binomial
)

summary(logit_fit)
```

```
##
## Call:
## glm(formula = severity ~ age + gender + race + smoking + bmi +
##     hypertension + diabetes + SBP + LDL + vaccine + depression,
##     family = binomial, data = train_dat)
##
## Coefficients:
##                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)       -23.391176   3.365722  -6.950 3.66e-12 ***
## age                 0.090661   0.027512   3.295 0.000983 ***
## genderMale         -0.015373   0.210266  -0.073 0.941715
```

```
## raceAsian           0.273030   0.470198   0.581 0.561463
## raceBlack          -0.065149   0.258748  -0.252 0.801206
## raceHispanic       -0.102350   0.343499  -0.298 0.765731
## smokingFormer       0.330010   0.233271   1.415 0.157154
## smokingCurrent      1.114327   0.333848   3.338 0.000844 ***
## bmi                 0.260503   0.043767   5.952 2.65e-09 ***
## hypertensionYes    -0.351095   0.350140  -1.003 0.315992
## diabetesYes         0.141854   0.288059   0.492 0.622402
## SBP                 0.099607   0.024021   4.147 3.37e-05 ***
## LDL                -0.008684   0.005437  -1.597 0.110243
## vaccineVaccinated  -3.527781   0.249279 -14.152  < 2e-16 ***
## depression         -0.071642   0.051379  -1.394 0.163200
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 997.66  on 798  degrees of freedom
## Residual deviance: 598.93  on 784  degrees of freedom
## AIC: 628.93
##
## Number of Fisher Scoring iterations: 5
```

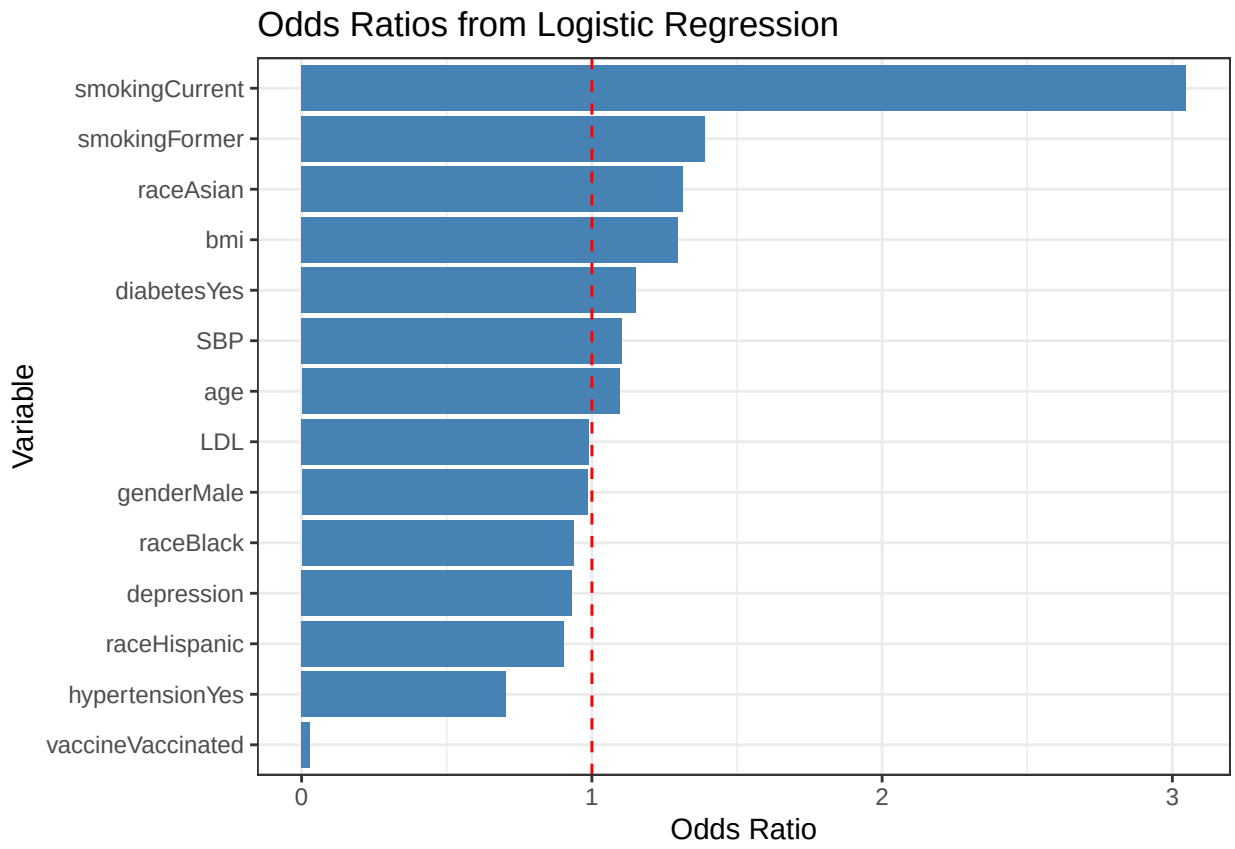
```r
# OR
exp(cbind(
  OR = coef(logit_fit),
  confint(logit_fit)
))
```

```
## Waiting for profiling to be done...
```

```
##                              OR        2.5 %       97.5 %
## (Intercept)       6.939707e-11 7.812562e-14 4.291448e-08
## age               1.094898e+00 1.037957e+00 1.156346e+00
## genderMale        9.847442e-01 6.510128e-01 1.486461e+00
## raceAsian         1.313940e+00 5.157043e-01 3.273127e+00
## raceBlack         9.369275e-01 5.619677e-01 1.552268e+00
## raceHispanic      9.027136e-01 4.560450e-01 1.757536e+00
## smokingFormer     1.390982e+00 8.804538e-01 2.200539e+00
## smokingCurrent    3.047516e+00 1.588231e+00 5.893067e+00
## bmi               1.297583e+00 1.193052e+00 1.416724e+00
## hypertensionYes   7.039169e-01 3.528559e-01 1.395435e+00
## diabetesYes       1.152409e+00 6.521443e-01 2.021667e+00
## SBP               1.104736e+00 1.054744e+00 1.159061e+00
## LDL               9.913535e-01 9.808131e-01 1.001979e+00
## vaccineVaccinated 2.937001e-02 1.767488e-02 4.705515e-02
## depression        9.308639e-01 8.410008e-01 1.029013e+00
```

```r
# Plot
or_df <- data.frame(
  Variable = rownames(coef(summary(logit_fit))),
  OR = exp(coef(logit_fit))
)
```

```r
or_df <- or_df %>%
  filter(Variable != "(Intercept)") %>%
  arrange(desc(OR))
ggplot(
  or_df,
  aes(
    x = reorder(Variable, OR),
    y = OR
  )
) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  geom_hline(
    yintercept = 1,
    linetype = "dashed",
    color = "red"
  ) +
  theme_bw() +
  labs(
    title = "Odds Ratios from Logistic Regression",
    x = "Variable",
    y = "Odds Ratio"
  )
```



```r
# Predicted Probabilities
logit_prob <- predict(
  logit_fit,
  newdata = test_dat,
  type = "response"
)
```

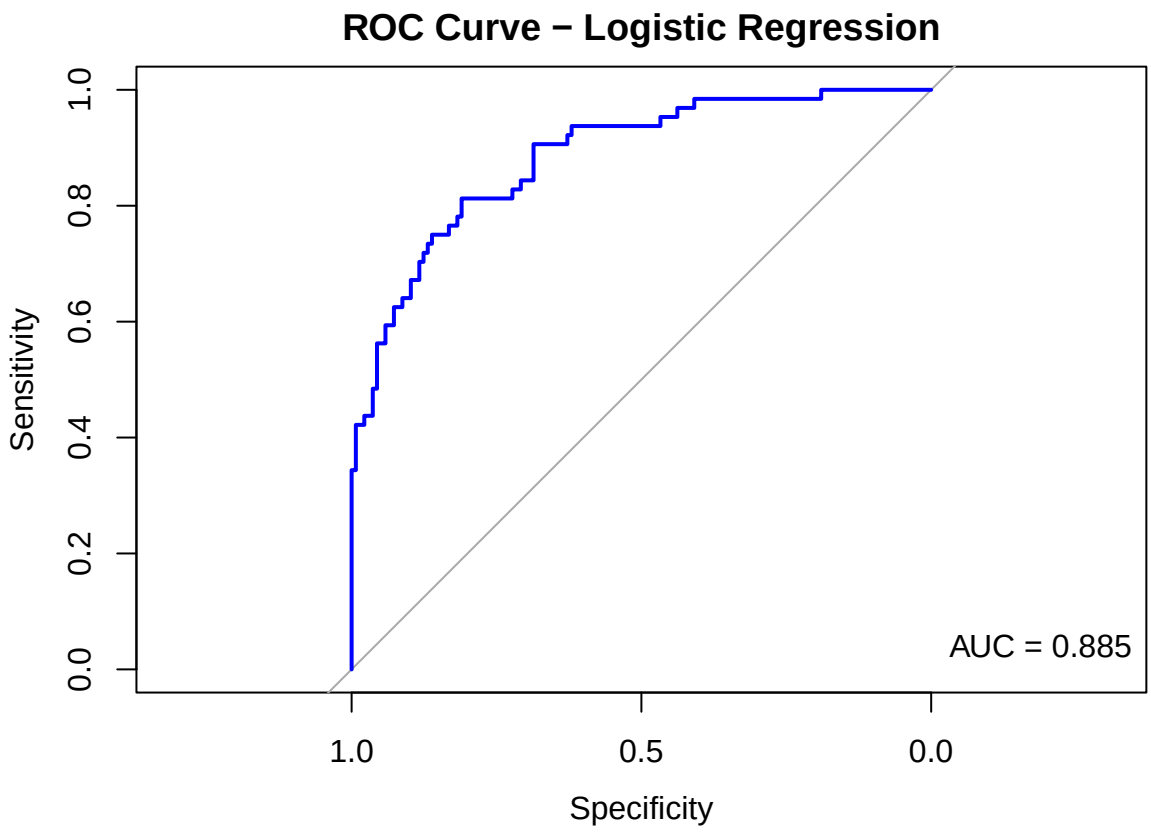
```r
# ROC
roc_logit <- roc(
  response = test_dat$severity,
  predictor = logit_prob,
  levels = c("NotSevere","Severe")
)
```

```
## Setting direction: controls < cases
```

```r
auc_logit <- auc(roc_logit)

plot(
  roc_logit,
  main = "ROC Curve - Logistic Regression",
  col = "blue",
  lwd = 2
)

legend(
  "bottomright",
  legend = paste("AUC =", round(auc_logit, 3)),
  bty = "n"
)
```



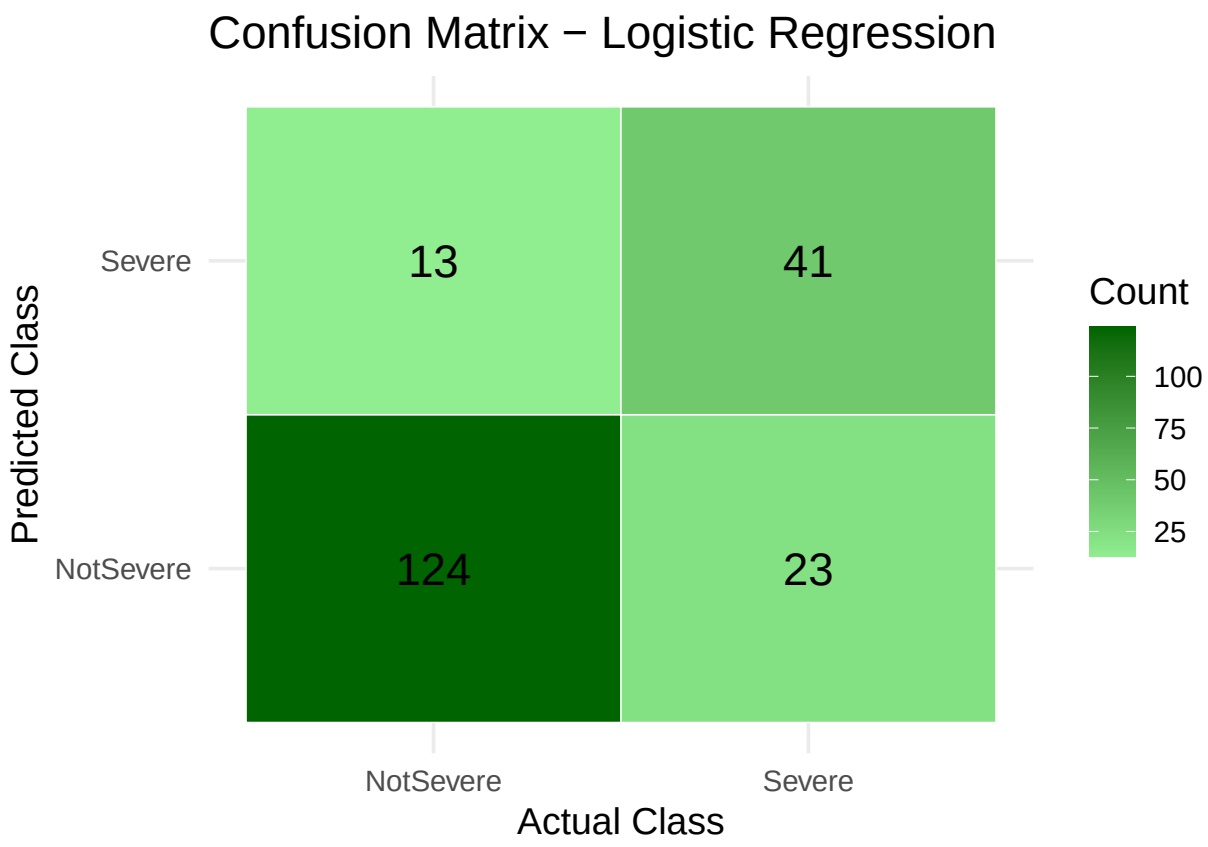

```
# Predictions
logit_pred <- ifelse(logit_prob > 0.5,
                     "Severe",
                     "NotSevere")

logit_pred <- factor(
  logit_pred,
  levels = c("NotSevere","Severe")
)

# Confusion matrix
cm_logit <- confusionMatrix(
  data = logit_pred,
  reference = test_dat$severity,
  positive = "Severe"
)

cm_logit_df <- as.data.frame(cm_logit$table)
# Plot
ggplot(cm_logit_df, aes(x = Reference, y = Prediction, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq), size = 6) +
  scale_fill_gradient(low = "lightgreen", high = "darkgreen") +
  labs(
    title = "Confusion Matrix - Logistic Regression",
    x = "Actual Class",
```

```
    y = "Predicted Class",
    fill = "Count"
  ) +
  theme_minimal(base_size = 14)
```



Boosting

```
ctrl <- trainControl(
  method = "cv",
  number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary
)
```

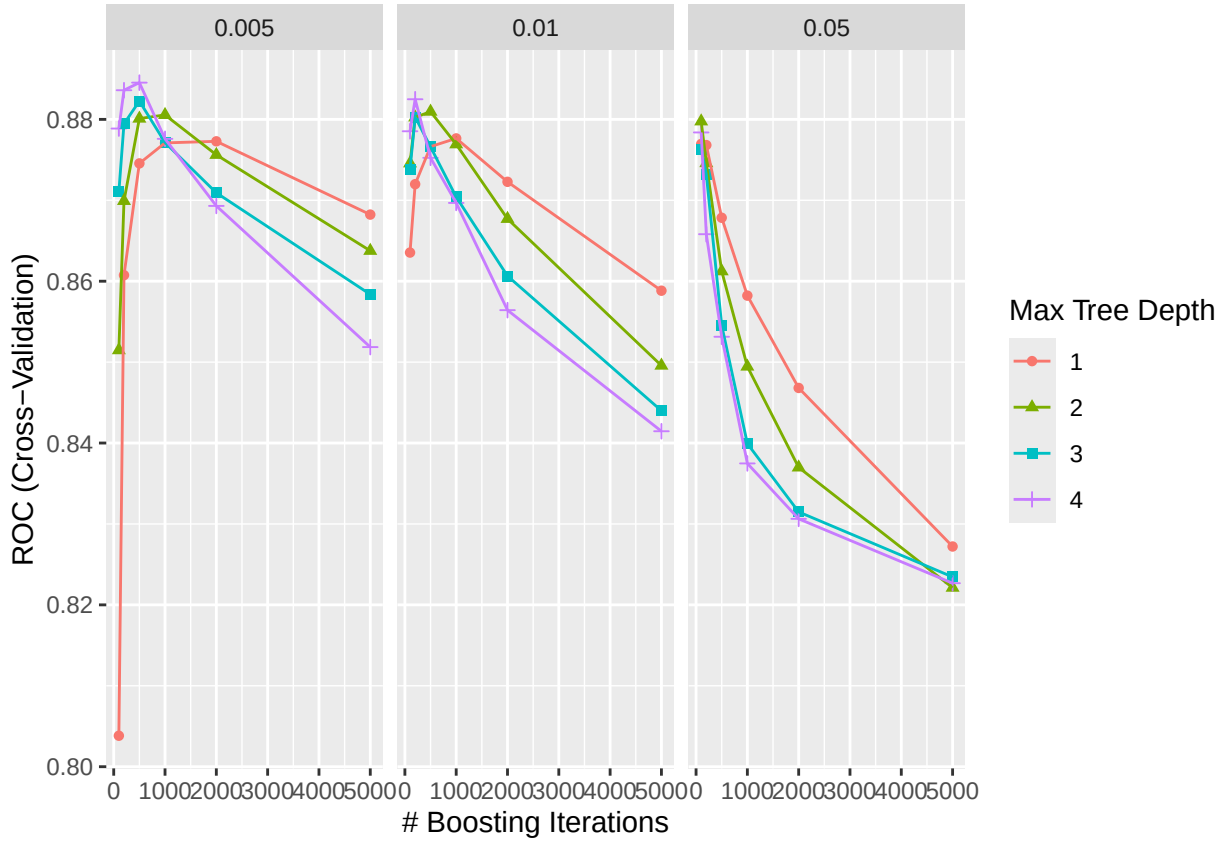
```
# tuning grid
gbm.grid <- expand.grid(
  n.trees = c(100, 200, 500, 1000, 2000, 5000),
  interaction.depth = 1:4,
  shrinkage = c(0.005, 0.01, 0.05),
  n.minobsinnode = 10
)
```

```r
# Train
set.seed(4027)

gbm_fit <- train(
  severity ~ .,
  data = train_dat,
  method = "gbm",
  metric = "ROC",
  tuneGrid = gbm.grid,
  trControl = ctrl,
  verbose = FALSE
)
```

```r
ggplot(gbm_fit)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## i The deprecated feature was likely used in the caret package.
##   Please report the issue at <https://github.com/topepo/caret/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



```r
# Best parameters
gbm_fit$bestTune
```

```
##    n.trees interaction.depth shrinkage n.minobsinnode
## 21     500                 4     0.005             10
```

```r
gbm_prob <- predict(
  gbm_fit,
  newdata = test_dat,
  type = "prob"
)[,"Severe"]
```

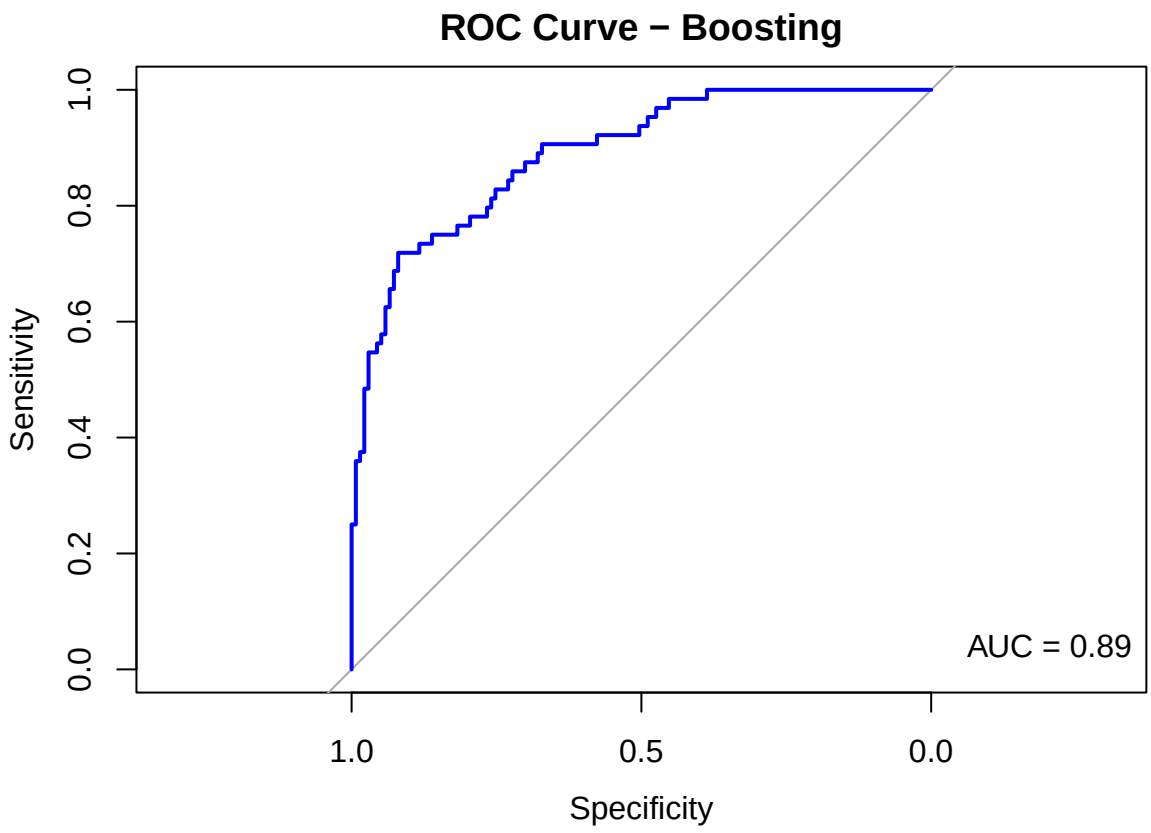
```r
roc_gbm <- roc(
  response = test_dat$severity,
  predictor = gbm_prob,
  levels = c("NotSevere","Severe")
)
```

```
## Setting direction: controls < cases
```

```r
auc_gbm <- auc(roc_gbm)

plot(
  roc_gbm,
  main = "ROC Curve - Boosting",
  col = "blue",
  lwd = 2
)

legend(
  "bottomright",
  legend = paste(
    "AUC =",
    round(auc_gbm, 3)
  ),
  bty = "n"
)
```



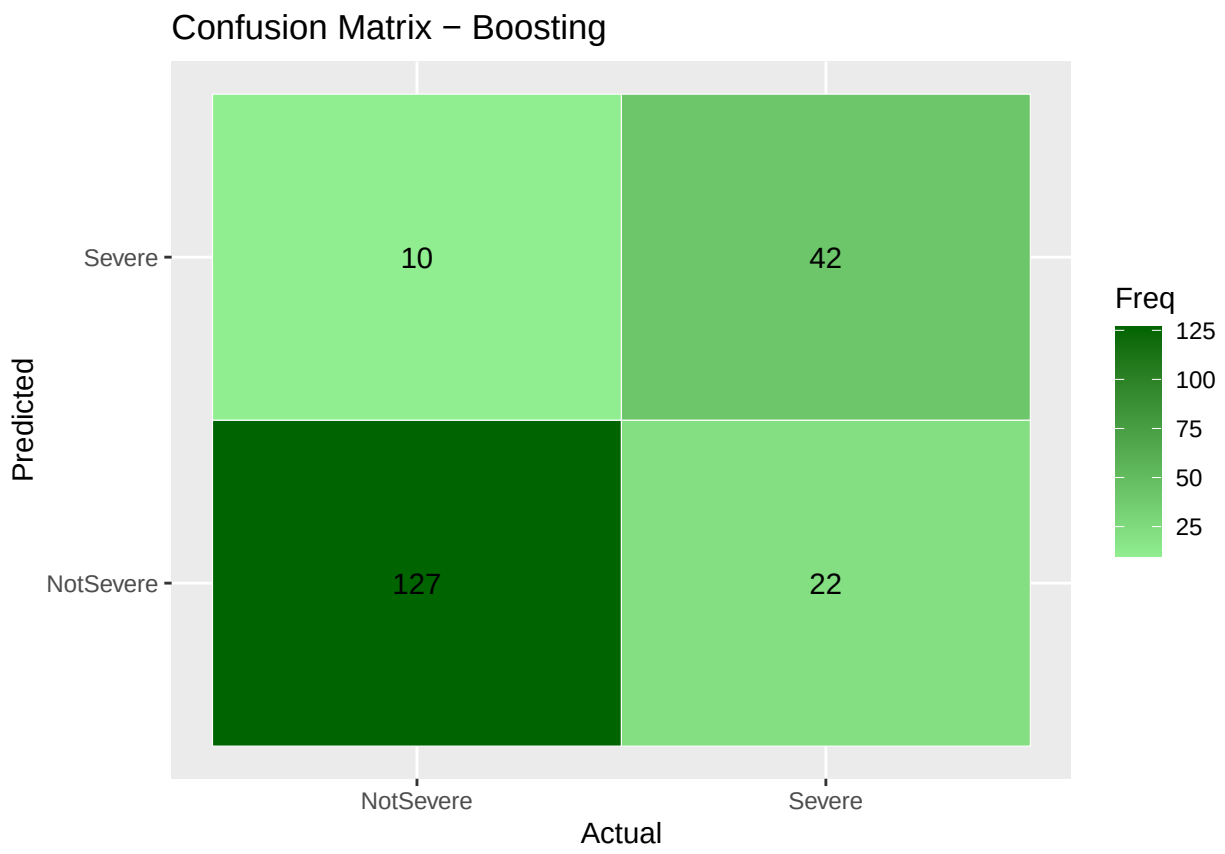

```
gbm_pred <- predict(
  gbm_fit,
  newdata = test_dat
)

cm_gbm <- confusionMatrix(
  data = gbm_pred,
  reference = test_dat$severity
)

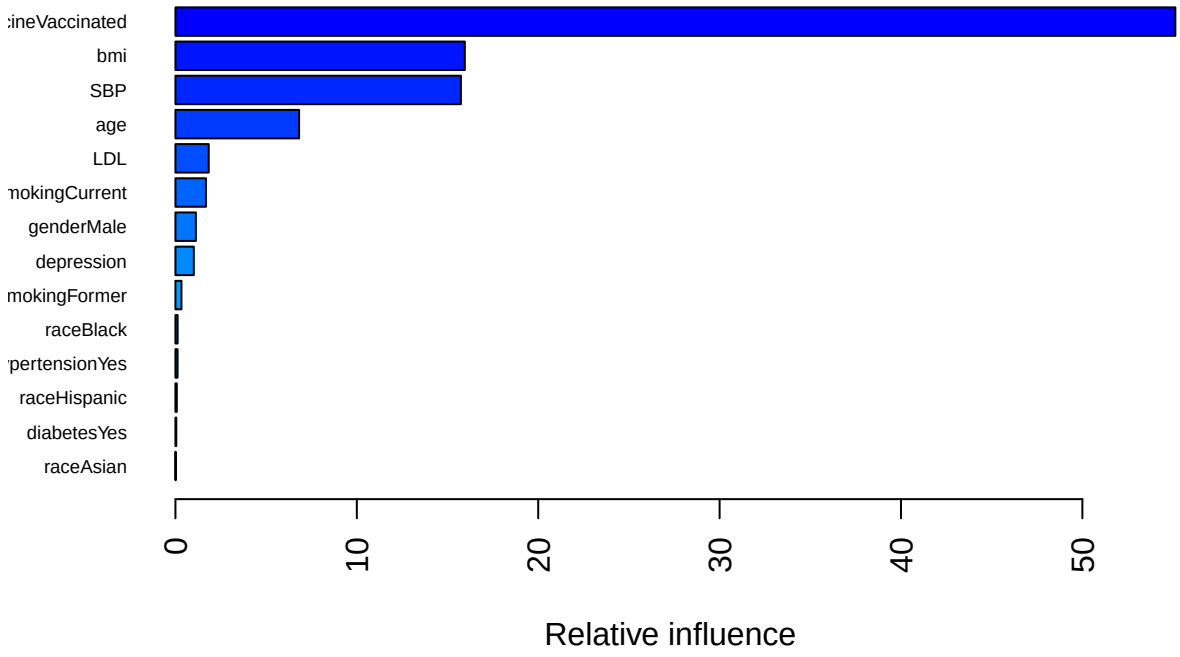
cm_gbm_df <- as.data.frame(cm_gbm$table)

ggplot(
  cm_gbm_df,
  aes(
    x = Reference,
    y = Prediction,
    fill = Freq
  )
) +
  geom_tile(color = "white") +
  geom_text(
    aes(label = Freq)
  ) +
  scale_fill_gradient(
    low = "lightgreen",
```

```
    high = "darkgreen"
  ) +
  labs(
    title = "Confusion Matrix - Boosting",
    x = "Actual",
    y = "Predicted"
  )
```




```
summary(gbm_fit$finalModel, las = 2, cBars = 19, cex.names = 0.6)
```




```
##                                 var     rel.inf
## vaccineVaccinated vaccineVaccinated 55.12600674
## bmi                             bmi 15.95420522
## SBP                             SBP 15.73816176
## age                             age  6.81860675
## LDL                             LDL  1.83682402
## smokingCurrent       smokingCurrent  1.68166773
## genderMale               genderMale  1.12610169
## depression               depression  1.01239074
## smokingFormer         smokingFormer  0.33257521
## raceBlack                 raceBlack  0.12053936
## hypertensionYes     hypertensionYes  0.11715920
## raceHispanic           raceHispanic  0.07435451
## diabetesYes             diabetesYes  0.04640105
## raceAsian                 raceAsian  0.01500604
```

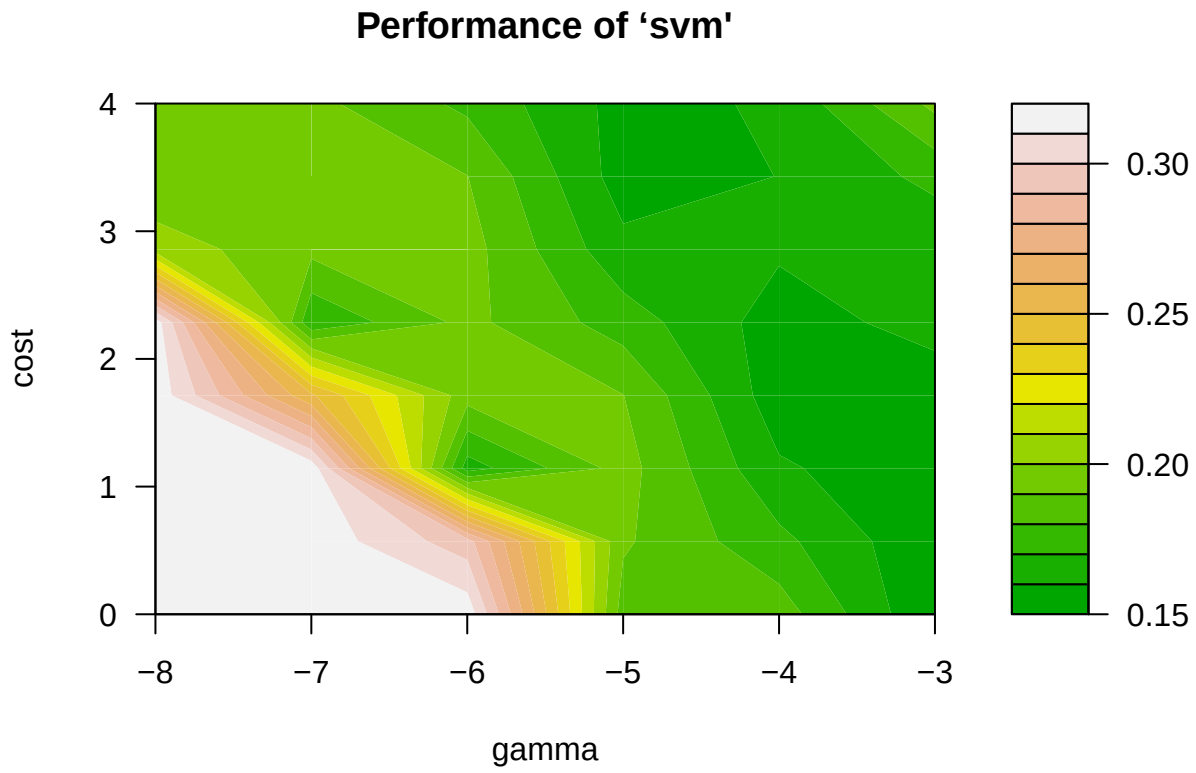
SVM

```
set.seed(4027)

radial.tune <- tune.svm(
  severity ~ .,
  data = train_dat,
```

```
  kernel = "radial",
  probability = TRUE,
  cost = exp(seq(0, 4, length = 8)),
  gamma = exp(seq(-8, -3, length = 6))
)
```

```
plot(
  radial.tune,
  transform.y = log,
  transform.x = log,
  color.palette = terrain.colors
)
```




```
radial.tune$best.parameters
```

```
##        gamma cost
## 6 0.04978707    1
```

```
best.radial <- radial.tune$best.model
summary(best.radial)
```

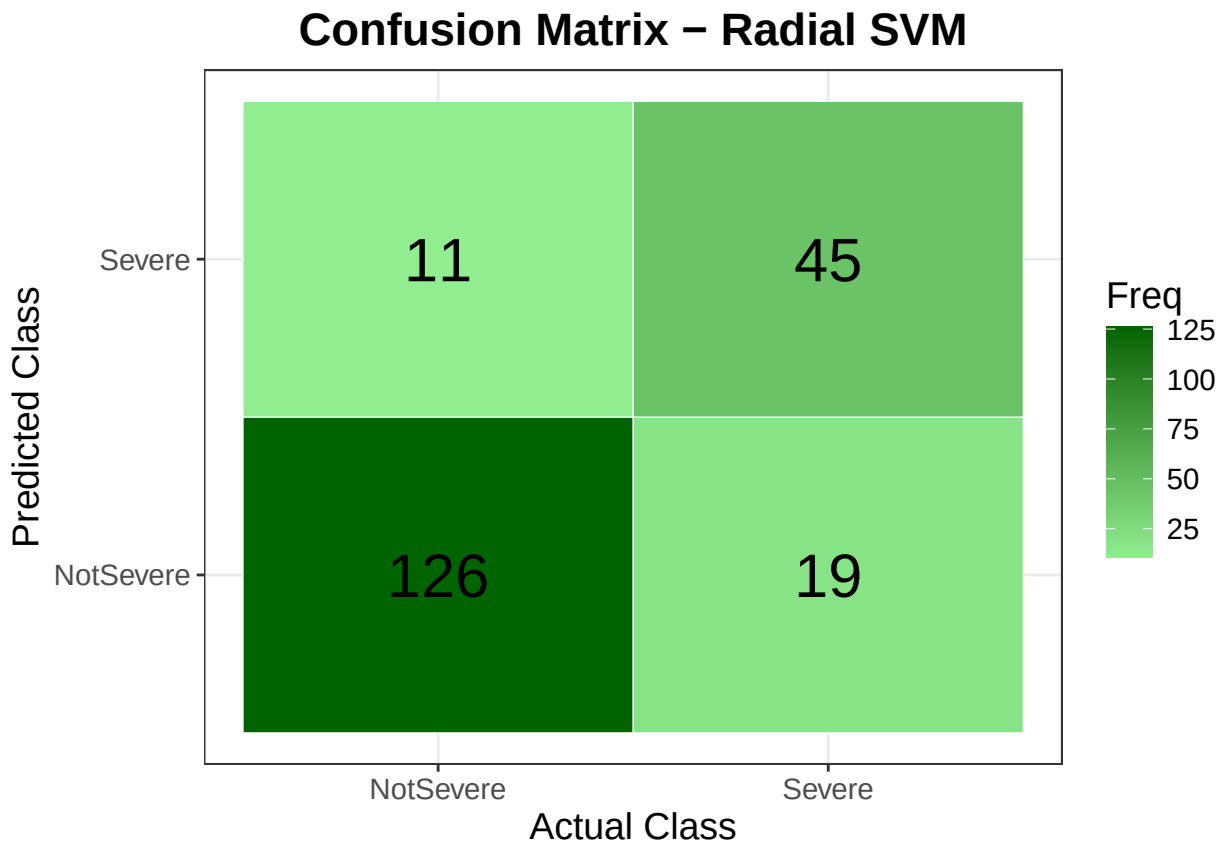
```
##
## Call:
## best.svm(x = severity ~ ., data = train_dat, gamma = exp(seq(-8,
```

```
##     -3, length = 6)), cost = exp(seq(0, 4, length = 8)), kernel = "radial",
##     probability = TRUE)
##
##
## Parameters:
##    SVM-Type:  C-classification
##  SVM-Kernel:  radial
##        cost:  1
##
## Number of Support Vectors:  394
##
##  ( 204 190 )
##
##
## Number of Classes:  2
##
## Levels:
##  NotSevere Severe
```

```
pred.radial <- predict(
  best.radial,
  newdata = test_dat
)
cm_svm <- confusionMatrix(
  data = pred.radial,
  reference = test_dat$severity
)
cm_svm_df <- as.data.frame(cm_svm$table)
colnames(cm_svm_df) <- c("Prediction", "Reference", "Freq")
ggplot(cm_svm_df,
       aes(x = Reference,
           y = Prediction,
           fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Freq),
            size = 8) +
  scale_fill_gradient(low = "lightgreen",
                      high = "darkgreen") +

  labs(
    title = "Confusion Matrix - Radial SVM",
    x = "Actual Class",
    y = "Predicted Class"
  ) +

  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    text = element_text(size = 14)
  )
```




```
# Predicted probabilities
svm_prob <- predict(
  best.radial,
  newdata = test_dat,
  probability = TRUE
)

svm_prob <- attr(
  svm_prob,
  "probabilities"
)[,"Severe"]
```

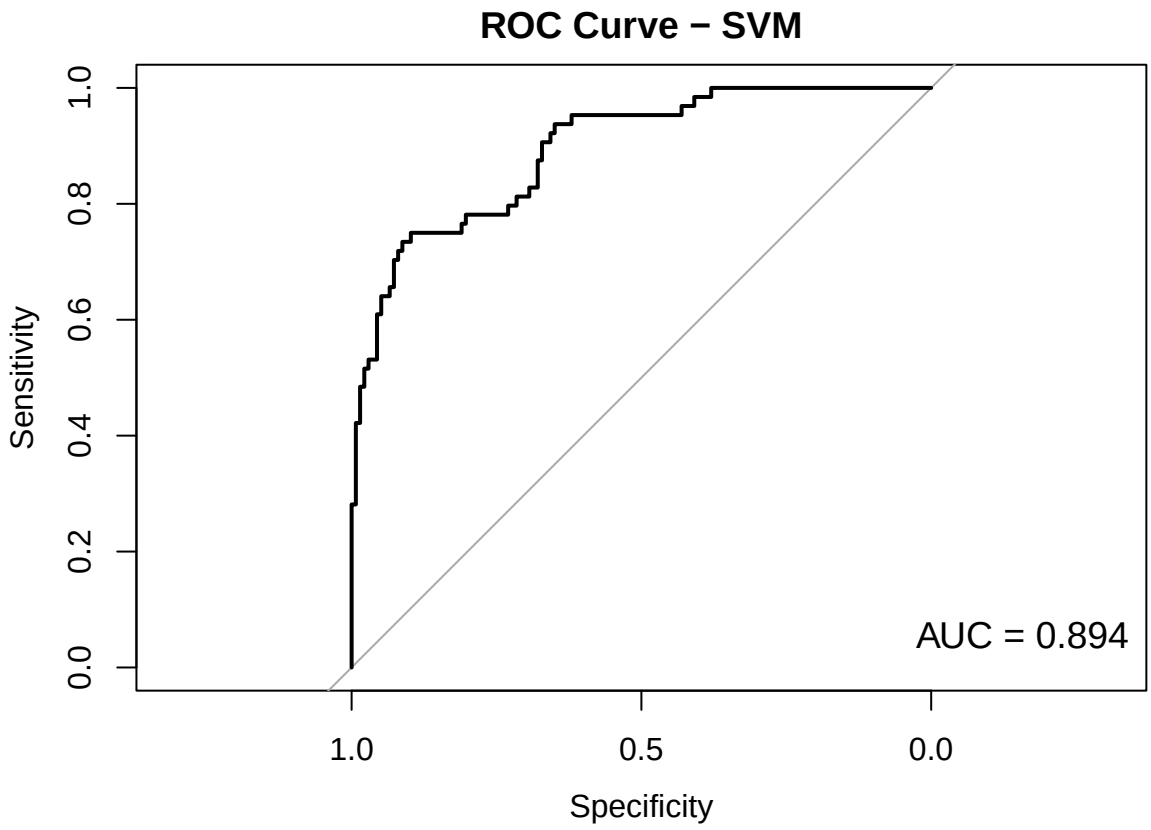
```
#ROC/AUC
roc_svm <- roc(
  response = test_dat$severity,
  predictor = svm_prob,
  levels = c("NotSevere","Severe")
)
```

```
## Setting direction: controls < cases
```

```
svm_auc <- auc(roc_svm)

plot(
  roc_svm,
```

```
  main = "ROC Curve - SVM"
)
legend("bottomright",  legend = paste("AUC =", round(svm_auc, 3)),  bty = "n",  cex = 1.2)
```




```
explainer_svm <- explain(
  model = best.radial,

  data = train_dat %>%
    select(-severity),

  y = as.numeric(train_dat$severity == "Severe"),

  label = "Radial SVM",

  verbose = FALSE
)

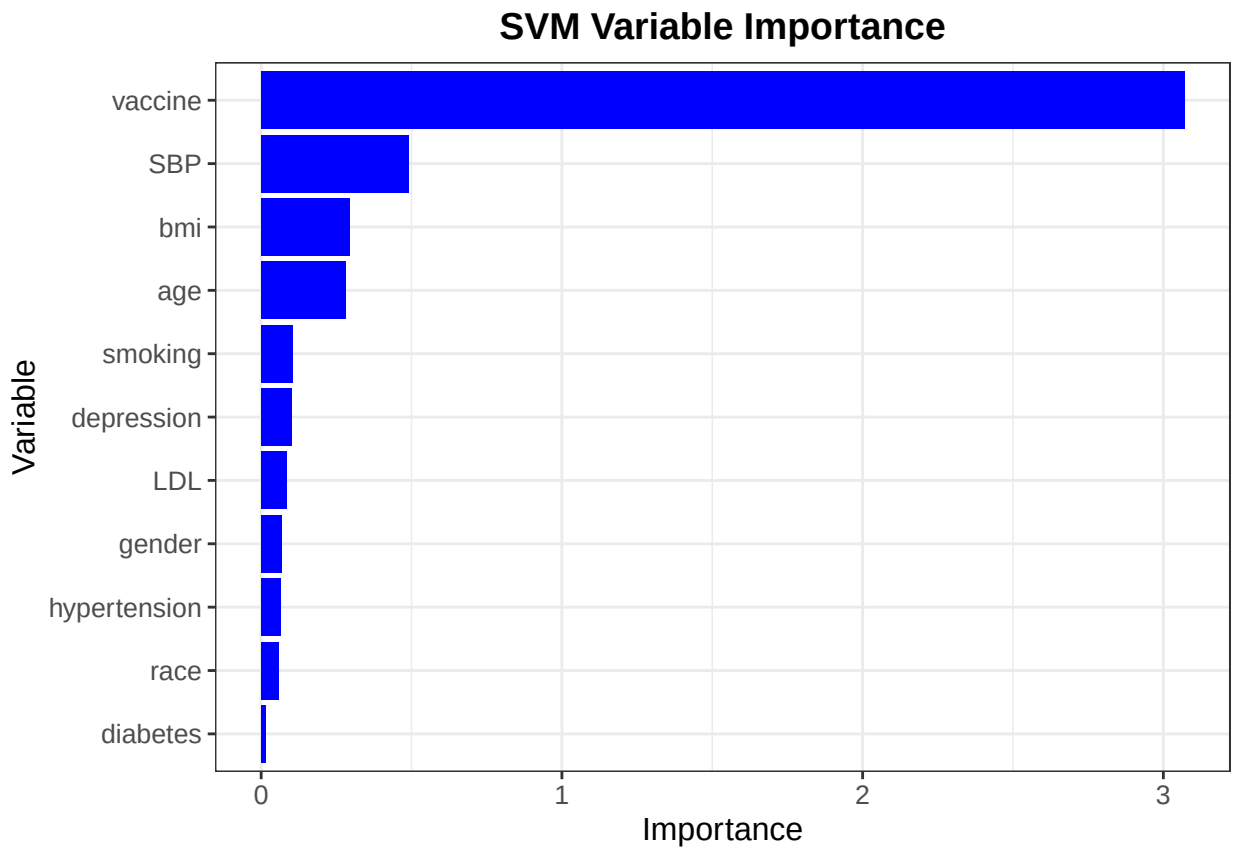

svm_vi <- model_parts(
  explainer_svm,
  type = "difference"
)


svm_vi_df <- svm_vi %>%
```

```
  as.data.frame()

svm_vi_df <- svm_vi_df %>%
  filter(
    variable != "_baseline_",
    variable != "_full_model_"
  )

ggplot(
  svm_vi_df,
  aes(
    x = reorder(variable, dropout_loss),
    y = dropout_loss
  )
) +
  geom_col(fill = "blue") +
  coord_flip() +
  theme_bw() +
  labs(
    title = "SVM Variable Importance",
    x = "Variable",
    y = "Importance"
  ) +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    ),
    text = element_text(size = 12)
  )
```



Model Comparison

```
# Model comparison table

comparison_table <- tibble(
  Model = c("Logistic Regression", "Boosting", "SVM"),

  Accuracy = c(
    cm_logit$overall["Accuracy"],
    cm_gbm$overall["Accuracy"],
    cm_svm$overall["Accuracy"]
  ),

  Sensitivity = c(
    cm_logit$byClass["Sensitivity"],
    cm_gbm$byClass["Sensitivity"],
    cm_svm$byClass["Sensitivity"]
  ),

  Specificity = c(
    cm_logit$byClass["Specificity"],
    cm_gbm$byClass["Specificity"],
    cm_svm$byClass["Specificity"]
```

```
  ),

  AUC = c(
    as.numeric(auc_logit),
    as.numeric(auc_gbm),
    as.numeric(svm_auc)
  )
)

comparison_table %>%
  mutate(across(where(is.numeric), round, 3)) %>%
  knitr::kable(caption = "Model Comparison")
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(where(is.numeric), round, 3)`.
## Caused by warning:
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
##   # Previously
##   across(a:b, mean, na.rm = TRUE)
##
##   # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

Table 1: Model Comparison

Model	Accuracy	Sensitivity	Specificity	AUC
Logistic Regression	0.821	0.641	0.905	0.885
Boosting	0.841	0.927	0.656	0.890
SVM	0.851	0.920	0.703	0.894